What Context Cannot Add, and What It Cannot Subtract

Geist Atlas · Module 19 · What the Tutor Needs to Know

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[**SCOPE-BOUND**] Supported within a narrow scope · **Family:** What the Tutor Needs to Know
Claim: The programme's negative results fall into two classes. Redundancy limits: context engineering cannot add information a capable model already extracts from the transcript, which is why learner models, profiles, standing advice, and composed prompt machinery returned nulls against lean baselines. Subtraction limits: prompt-level means could not remove conduct that training installed — a tutor asked to withhold the warrant of its own frame delivered that conduct in 0 of 8 dialogues under instruction and survived structural repair in 0 of 9, writing a new justification each time the violating sentence was quoted back.

A discussion-tier synthesis, not a new measurement. Prompt-level adaptation selects among completions the model already affords; it does not construct new capacities and it does not delete trained ones. The practical consequence for designers: an arm that needs the model to be less forthcoming than its training should be verified for delivered conduct before scoring, or declared unavailable. External work on preference training, negated instructions, and persona drift places the boundary where these studies found it.

Derived view of `paper-full-2.0.md` v3.0.297 — §7.16. The paper is canonical; edit it there, not here.

*This sentence is the only one actually authored by Liam Magee in this paper.

7.16 The limits of prompt-level adaptation: what context cannot add, and what it cannot subtract

The programme's negative results now outnumber its positive ones, and they are not a miscellany. The boundary map of §7.12 arranged them on an information-versus-enforcement ladder; this section states the underlying regularity and connects it to the external literature. Every intervention in this paper that does not change model weights — prompt, persona, conduct card, scaffold, gate — acts on the same object: the model's conditional distribution, through context. Context can re-weight completions the trained policy already supports; it cannot manufacture support that training removed, and it adds nothing by restating what the model already infers. Read this way, the accumulated nulls sort into exactly two classes.

Redundancy limits: context cannot add what the model already has. This is the class §7.11 derived for instructions ("instructions converge, signal accumulates") and the programme's repeated theory-of-mind results measured for scaffolds: the state-schema and bilateral-ToM ablations (§6.8.5–.6), the concealed-interior signal axis (§6.10), and the quasi-logical stall-watcher (§6.13.6), which found the deficit its arithmetic would correct never occurs — a strong learner voices every derivable fact at latency 0. The same shape recurs one level up wherever the added layer re-encodes what the forward pass conditions on anyway: the fitted `state → action` policy has no off-family headroom over the implicit base (§6.9.8); longitudinal character-state routing buys a modest efficiency gain, not a qualitative rescue, once the real learner is competent (§6.8.9); coached consolidation into durable memory changes no behaviour (§6.16); composition of validated mechanisms is sub-additive (§6.14); and in the adversarial-game probe the bottleneck was information *format*, never missing information (§6.15). The external literature carries the same finding at training grain: instruction tuning largely selects a format and behaviour distribution the pretrained model already holds, rather than adding capability (Zhou et al., 2023); elaborate agent scaffolds routinely fail to beat simple baselines once cost and evaluation are controlled (Kapoor et al., 2024); and machine theory-of-mind competence is fragile to trivial perturbation, which cautions against building layers on it (Ullman, 2023). The programme's internal verdict — adaptation gains come from routing *new* signal, not re-encoding inferable signal — is the local form of this class.

Subtraction limits: context cannot remove what training installed. The defiant-warrant close (§6.30) makes this class sharp because it holds the target conduct fixed and escalates the means. Asking the model to withhold the warrant of its own frame failed as instruction (0 of 8 delivered, with the card injected every turn) and then failed under structural enforcement with the violating sentence quoted back in the repair (0 of 9 survived): the tutor wrote a new justification each time. Enforcement could stop the turn but could not produce the conduct — the gate yielded typed refusals-to-score, not withholding. The stall-watcher null (§6.13.6) marks the same boundary from the observation side, not as a second live attempt: that design needed a learner that leaves derivable facts unvoiced, and the strong pinned learner never does — a measured floor, with the one lever that might have created the deficit (a weaker learner) deliberately left unpulled. The asymmetry against the additive designs is what carries the information: hostile registers, resistant personas, tutor moves, and warrant gates — conduct *added* to the default — could all be fielded and measured

(§6.17, §6.25–§6.28; imperfectly where worlds were hard, a delivery profile §6.28 reports as a result), while the two designs that required conduct to be *subtracted* from the default could not field their arm at all. The external literature explains why the boundary sits there. Preference training collapses output diversity around the preferred mode, removing precisely the alternative completions a subtractive instruction would need to select (Kirk et al., 2024); models follow affirmative instructions far better than negated ones, and on some negated tasks performance *worsens* with scale (Jang et al., 2022; McKenzie et al., 2023); trained-in accommodation to the interlocutor is a documented attractor (Sharma et al., 2024); and even successfully installed instructions decay back toward the default persona over dialogue turns (Li et al., 2024). On a strongly aligned stack, "serve the warrant" rides the trained current; "withhold it" swims against it.

One statement of the mechanism. Prompt-level adaptation is *selection*, not construction and not deletion. A context shifts probability mass among behaviours with support under the trained policy; where alignment training has concentrated that support — the justified answer, the voiced derivation, the served warrant — the complement has no mass to shift onto, and the strongest available enforcement converts the missing arm into refusal rather than compliance. This is also why the boundary map's ladder (§7.12) has the shape it has: the interventions that worked either selected supported behaviour (moves, registers, personas) or removed behaviour *from the pipeline* by construction (typed gates, harness-owned checks), and the one arc that changed weights rather than context — Program-2's trained warrant move (§6.20–§6.22) — is the one where a target conduct became newly deliverable and survived transfer. Subtraction, where it is needed, appears to live at the parameter level, consistent with §7.12's closing hypothesis that the missing adaptive capacity is parametric.

A design consequence. Some control arms are not implementable on strong aligned stacks, and a registration should treat them as unavailable rather than assume instruction will produce them. A design whose contrast requires the model to be less justifying, less forthcoming, or less helpful than its training should either verify delivered conduct before scoring (the §6.30 v2 gate is the reusable pattern: adjudicated delivery with quoted evidence, typed never-scored stops), relocate the subtracted conduct into the harness while acknowledging that this changes the object of study, or not run the arm. The silent alternative — scoring an arm the model never delivered — is the v1 failure mode, and it produces contrasts between labels, not behaviours.

Status. Interpretive synthesis of results reported in the sections cited; it introduces no empirical claim beyond §6.30's, and the external citations support interpretation, not new measurement.

Neighbouring modules: Module 7 — A Bigger Learner Model Did Not Help; Module 12 — More Layers Do Not Mean More Benefit; Module 14 — Advice, Enforcement, and the Cost of Control; Module 18 — Reaching the Learner Who Resists.

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